

Ex2: Bias Audit of Pre-Trained Word Embeddings Using the Word Embedding Association Test (WEAT)

Learning Objective:

Develop a program that performs a bias audit using Word Embedding Association Test (WEAT), to identify and quantify gender or racial stereotypes present in Google News Word2Vec embeddings.

Steps:

1. Import necessary Python libraries: `gensim`, `numpy`, etc.
2. Load the Google News Word2Vec embeddings. (2GB Dataset, takes time)
3. Define Target and Attribute Word Sets
 - a. Target words (X and Y): Groups to compare, e.g.,
 - i. Gender: `X = ["man", "male", "boy".....]`,
`Y = ["woman", "female", "girl".....]`
 - b. Attribute words (A and B): Concepts to test association with, e.g.,
 - i. Career vs Family: `A = ["career", "business".....]`,
`B = ["family", "home".....]`
4. Define Helper Functions – association function, weat score function, effect size function and compute weat score, effect size.
5. Interpret the Results as
 - a. Effect Size > 0: Target group X is more associated with A (bias towards X-A).
 - b. Effect Size < 0: Target group Y is more associated with A (bias towards Y-A).
 - c. Effect Size $\approx$ 0: No detectable bias.

Optional Extensions

1. Perform multiple WEAT tests for different bias types (**gender, racial, religious**).
2. Visualize effect sizes using bar charts for comparison.

Program:

```
import numpy as np
import gensim.downloader as api
from gensim.models import KeyedVectors
import os

MODEL_PATH = "GoogleNews-vectors-negative300.bin"

print("Attempting to load Google News Word2Vec model...")

if os.path.exists(MODEL_PATH):
    model = KeyedVectors.load_word2vec_format(
        MODEL_PATH,
        binary=True
    )
    print("Loaded Google News Word2Vec.\n")
```

```

else:
    print("Google News Word2Vec not found.")
    print("Loading GloVe Wiki Gigaword 300 instead (fallback)...\n")
    model = api.load("glove-wiki-gigaword-300")
    print("GloVe model loaded successfully.\n")

X = ["man", "male", "boy", "father", "brother", "son"]
Y = ["woman", "female", "girl", "mother", "sister", "daughter"]

A = ["career", "business", "corporation", "salary", "office", "professional"]
B = ["family", "home", "children", "parent", "marriage", "house"]

def filter_vocab(words):
    return [w for w in words if w in model.key_to_index]

X = filter_vocab(X)
Y = filter_vocab(Y)
A = filter_vocab(A)
B = filter_vocab(B)

print("Final Word Sets Used:")
print("X (Male) :", X)
print("Y (Female) :", Y)
print("A (Career) :", A)
print("B (Family) :", B)
print()

def cosine_similarity(w1, w2):
    return np.dot(model[w1], model[w2]) / (
        np.linalg.norm(model[w1]) *
        np.linalg.norm(model[w2])
    )

def association(w, A, B):
    return (
        np.mean([cosine_similarity(w, a) for a in A]) -
        np.mean([cosine_similarity(w, b) for b in B])
    )

def weat_score(X, Y, A, B):
    return (

```

```

        sum(association(x, A, B) for x in X) -
        sum(association(y, A, B) for y in Y)
    )
def effect_size(X, Y, A, B):
    assoc_X = np.array([association(x, A, B) for x in X])
    assoc_Y = np.array([association(y, A, B) for y in Y])

    mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
    pooled_std = np.std(
        np.concatenate([assoc_X, assoc_Y]),
        ddof=1
    )

    return mean_diff / pooled_std

score = weat_score(X, Y, A, B)
effect = effect_size(X, Y, A, B)

print("WEAT Bias Audit Results")
print("-----")
print(f"WEAT Score : {score:.4f}")
print(f"Effect Size d : {effect:.4f}")
print()

print("Interpretation:")
if effect > 0.1:
    print("Bias detected: Male-related words are more associated with Career.")
elif effect < -0.1:
    print("Bias detected: Female-related words are more associated with Career.")
else:
    print("No significant gender bias detected.")

```

Output:

```
*** Attempting to load Google News Word2Vec model...
Google News Word2Vec not found.
Loading GloVe Wiki Gigaword 300 instead (fallback)...

[-----] 100.0% 376.1/376.1MB downloaded
GloVe model loaded successfully.

Final Word Sets Used:
X (Male) : ['man', 'male', 'boy', 'father', 'brother', 'son']
Y (Female) : ['woman', 'female', 'girl', 'mother', 'sister', 'daughter']
A (Career) : ['career', 'business', 'corporation', 'salary', 'office', 'professional']
B (Family) : ['family', 'home', 'children', 'parent', 'marriage', 'house']

WEAT Bias Audit Results
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WEAT Score : 0.3415
Effect Size d : 0.6663

Interpretation:
Bias detected: Male-related words are more associated with Career.
```

Learning Outcome:

Students will be able to process word embedding datasets, WEAT was implemented to measure associations between target and attribute word sets, interpreted effect sizes to detect bias, and critically evaluated the presence of gender or racial stereotypes in pre-trained embeddings.